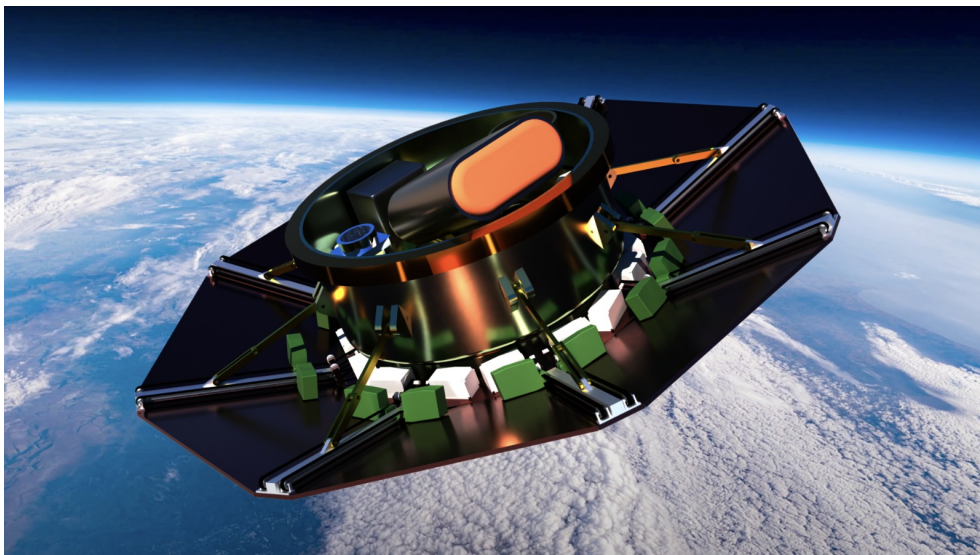


FOLDABLE AEROSHELL DEMONSTRATOR

Space HAVEN: Design and Dynamic Modelling of a Parachute Recovery System

FAD06 - Demonstrator Subsystem Design



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Abstract

This report describes the high-level design of a parachute recovery system for a small-scale foldable aeroshell, intended for use as a demonstrator for proof-of-concept of Imperial College London's HATHOR. The mission, Space HAVEN, involves a sub-orbital launch to an apogee of ≈ 320 km in order to reach hypersonic speeds on reentry. Parachutes were deemed necessary to slow the final phase of descent and reduce the ground impact velocity. The recovery system has been designed to be robust against deployment at any orientation, as the aerodynamic properties of the demonstrator at low speeds are not well-known and the demonstrator may tumble or fall inverted. To aid with this design, a 3-DOF model has been constructed to characterise the dynamic response of the demonstrator to parachute deployment. The model accounts for inflation dynamics of the parachute, and line-body interactions which will be present if the demonstrator tumbles or falls inverted. Key loading data can be extracted from the model, and has been shown to be capable of constructing an initial deployment envelope for the parachute subject to structural constraints. The model can be used in conjunction with structural design of the demonstrator in future design iterations, to ensure that the demonstrator is able to withstand the loads during the recovery sequence. Additionally, a CAD model has been created in order to visualise the demonstrator and extract inertial properties, along with a mass and power budget. The CAD model also verifies that the origami folding pattern used to fold the TPS panels is not statically determinate.

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List of Symbols, Subscripts and Abbreviations

Symbol	Description	Unit
a	Inertial/acceleration force	G
C_D	Drag coefficient	
C_L	Lift coefficient	
C_X	Opening force coefficient	
D	Drag	N
d	Diameter	m
g	Gravitational acceleration (9.81)	m/s ²
$I_{z'z'}$	Moment of inertia in radial axis	kgm ²
L	Lift	N
M	Mach number	
m	Mass	kg
r	Parachute line segment length	m
S	Parachute reference area	m ²
T	Parachute line tension	N
T_1	Induced force acting on payload fairing	N
T_2	Induced force acting on panel rim	N
t	Time	s
t_f	Parachute filling/opening time	s
V	Velocity	m/s
x	Vertical coordinate/altitude (Earth-fixed axis convention)	m
x'	Longitudinal coordinate (body-fixed axis convention)	m
y	Horizontal coordinate/range (Earth-fixed axis convention)	m
y'	Lateral/radial coordinate (body-fixed axis convention)	m
z'	Out-of-plane lateral/radial coordinate (body-fixed axis convention)	m
α	Angle of attack	deg
α_1	Payload fairing intersection angle of attack	deg
α_2	Panel rim intersection angle of attack	deg
β	Parachute line angle to vertical x -axis	deg
γ	Flight path angle	deg
δ	Angle between parachute and free-stream line (for $\alpha_1 < \alpha < \alpha_2$)	deg
ε	Angle under parachute line at payload fairing (for $\alpha_1 < \alpha < \alpha_2$)	deg
ε_1	Angle under parachute line at payload fairing (for $ \alpha > \alpha_2$)	deg
ζ	Angle between T_1 and x' -axis (for $\alpha_1 < \alpha < \alpha_2$)	deg
ζ_1	Angle between T_1 and x' -axis (for $ \alpha > \alpha_2$)	deg
η	Angle under parachute line at panel rim	deg
θ	Pitch angle	deg
μ	Angle between parachute and free-stream line (for $ \alpha > \alpha_2$)	deg
ξ	Angle between T_2 and x' -axis (for $ \alpha > \alpha_2$)	deg
ρ	Air density	kg/m ³
σ	Angle between x' -axis and payload-rim line (for $ \alpha > \alpha_2$)	deg
Subscript	Description	
CG	Centre of gravity	
CP	Centre of pressure	
chu	Parachute	
h	HATHOR/demonstrator	
PL OR p	Payload fairing	
RIM OR r	Rim of demonstrator panels	

0	Initial value
∞	Free-stream value

Abbreviation	Description
ADEPT	Adaptable Deployable Entry and Placement Technology
CAD	Computer Aided Design
CG	Centre of Gravity
CP	Centre of Pressure
CIRA	<i>Centro Italiano Ricerche Aerospaziali</i> (Italian Aerospace Research Centre)
DOF	Degree of Freedom
HATHOR	Hypersonic foldable Aeroshell for THERmal protection using ORigami
Space HAVEN	Space Hypersonic Aeroshell Validation ExperimeNt
NASA	National Aeronautics and Space Administration
TPS	Thermal Protection System
TRL	Technology Readiness Level

1 Introduction and Mission Overview

HATHOR is a foldable aeroshell concept designed to enable heavier payloads to land on planets and moons with atmospheres, such as Mars and Titan. The origami folding pattern allows it to be stowed in a small diameter launch platform and deploy to its full diameter shortly before atmospheric entry, providing drag to slow its descent [1]. To improve its Technology Readiness Level (TRL) and begin to validate the simulations and predictions that have been made in ongoing research, it is necessary to conduct a test flight of a small-scale demonstrator on Earth. Thus a model has been designed to fit into a sounding rocket and be deployed in a sub-orbital test flight. The Space HAVEN mission profile after ejection from the launch vehicle consists of four major segments, as shown in Figure 1. Phase A consists of reorientation after ejection and attitude control using reaction wheels [2], and deployment of the foldable panels to its fully open configuration near apogee (≈ 320 km) [3][4]. Phase B will commence once the demonstrator descends into the upper levels of the mesosphere (≈ 85 km), and involves guided reentry using a moving mass system [5]. Phase C will be an uncontrolled and unpredictable portion of flight where the demonstrator may tumble or be otherwise unstable. After collecting data on this portion for a sufficient period of time, a parachute will be deployed at the start of Phase D to enable a soft touch-down. Telemetry will be continuously streamed to a ground station and be stored on on-board data storage units, in order to analyse and characterise the flight after landing [6].

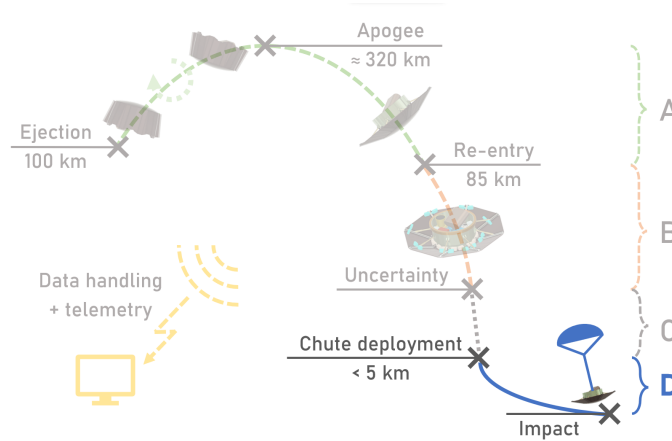


Figure 1: Space HAVEN mission overview

This report will aim to characterise Phase D and the dynamics of parachute deployment, and justify the use and implementation of a parachute recovery system. A 3-Degree-of-Freedom (DOF) model has been constructed to aid this analysis, and extraction of key loading information from the model has enabled a preliminary parachute deployment envelope to be constructed. Work has also been done on preparing mass and power budgets (see Appendix A) and creating a Computer-Aided Design (CAD) model of the demonstrator (see Appendix B), to aid with system placement and inertial tensor estimation, as well as visual verification of the origami folding pattern.

2 Justification

Since HATHOR is designed to produce high drag, it naturally will have a relatively low ballistic coefficient compared to typical reentry vehicles or other payloads on sounding rocket missions. Therefore it is not immediately necessary to use a parachute - similar test vehicles such as NASA's Nano-ADEPT [7] and CIRA's Mini-IRENE [8] elected to avoid using parachutes. However these vehicles had ballistic coefficients of 20 and 25 kg/m² respectively, whereas the Space HAVEN demonstrator is estimated to have a ballistic coefficient around 50 kg/m² based on current mass and drag estimates, and is significantly heavier than both Nano-ADEPT and Mini-IRENE at 45 kg. This is largely due to the decision to include active control and guidance systems both outside the atmosphere and within. It therefore

would have a high ground impact velocity of 28 m/s, risking significant damage to the structure and the payload, meaning a potential loss of key information regarding structural health, TPS health, and on-board data logging. Furthermore Nano-ADEPT underwent extensive ground-hit testing of key components before its flight [9], which has not been considered or budgeted for in this mission. The idea of actively inverting HATHOR in order to make use of its umbrella-like shape was also considered, but the drag coefficient increase was predicted to be insufficient to produce any significant reduction in ground impact velocity. Modelling and simulating the aerodynamic properties of the aft-body geometry was also deemed to be overly complex given the short project duration. Therefore the use of parachutes is a sensible and common solution to aid with the terminal portion of descent. Furthermore, future HATHOR missions at a larger scale delivering payloads to Mars would likely require a parachute, due to the thin atmosphere, so developing the techniques of implementing and modelling parachutes on a deployable aeroshell at this scale provides a good level of future-proofing and early validation.

3 Implementation

3.1 Parachute Selection and Sizing

There are dozens of parachute types in use for spacecraft applications, each with their own ideal use case envelopes and design points. Key properties of the parachute that need to be considered are its diameter and shape. The shape will inform the drag coefficient, stability properties, and suitable Mach/speed ranges for the chute to be used in. Tables 5-1 to 5-5 in Reference [10] gives these properties for a number of parachutes. The search for an appropriate parachute can first be narrowed down by selecting axisymmetric parachutes, as a guided descent with lifting parachutes is not needed. The low Mach number during terminal descent negates the need for a supersonic parachute. Therefore a conical ribbon parachute has been selected, as it has excellent stability properties, with a reasonable drag coefficient approximately in the middle of the range of alternatives, and a low predicted opening force coefficient. These properties are shown in Table 1, extracted from Table 5-2 in Reference [10].

Table 1: Typical properties of a conical ribbon parachute

Plan	Profile	Drag Coef. ($C_{D_{chu}}$)	Opening Force Coef. (C_X)	Avg. Oscillation Angle (°)	General Application
		0.50 to 0.55	≈ 1.05	0 to ± 3	Descent, Deceleration, $0.1 < M < 2.0$

Furthermore, simple empirical relations exist for the opening behaviour of ribbon parachutes [10], which makes them simple to implement in the Level B dynamic model (discussed in Section 4.2).

The parachute can be sized based on the desired ground impact velocity. Table 5.12.1-1 in NASA's Sounding Rocket handbook [11] gives suggested parachute types and sizes for varying payload masses, from which it can be estimated that a ground impact velocity of 9-10 m/s is recommended. Since the demonstrator payload bay is very limited in volume, however, a ground impact velocity of 12 m/s has been selected in order to slightly decrease the packed volume requirement of the parachute. The desired diameter can therefore be calculated as follows:

$$d_{chu} = \sqrt{\frac{1}{C_{D_{chu}}} \left[\frac{8m_h g}{\rho V^2 \pi} - d_h^2 C_{D_h} \right]} \quad \text{where:} \quad \begin{aligned} C_{D_{chu}} &= 0.55 \\ C_{D_h} &= 1.1 \\ d_h &= 0.96 \text{ m} \\ m_h &= 45 \text{ kg} \end{aligned} \quad (1)$$

This gives a required diameter d_{chu} of 3.12 m. The parachute suspension lines should typically be 1 to

2 times greater in length than the parachute diameter [12] [13], however since the demonstrator is a high-drag body with no aerodynamic back-shell, the wake is likely to be particularly turbulent and intense, therefore a preliminary line length of 8 m ($2.56 \times d_{chu}$) has been selected.

3.2 Deployment System

The recovery system for the demonstrator needs to be able to safely deploy the parachute at any orientation. Typical parachute recovery systems are designed for vehicles with a known orientation during descent, however the uncertainty surrounding the demonstrator's transonic and subsonic behaviour means the descent orientation cannot yet be estimated. A deployment system has therefore been designed which will be robust against almost all deployment orientations. It makes use of a mortar system, automatically deployed at a predetermined altitude by the flight computers, as this is a common and widely tested deployment method suitable for a wide range of Mach numbers [14].

Figure 2 shows a high-level deployment system concept inside the payload section of the demonstrator. Sufficient room has been left underneath the stadium-shaped portion for mortar hardware, ejection charges, and other components necessary for deployment. When triggered, the mortar will eject the parachute (shown in orange) radially outwards and 20° above the body horizontal axis. A parachute bag will ensure the parachute remains closed until after the lines become taut, reducing initial shock loads. The top panel of the system will be hinged in order to allow the parachute lines to extend vertically upwards from the centre of the demonstrator when inflated. Packed volume estimates have been inferred from a similar-sized parachute for small rockets [15].

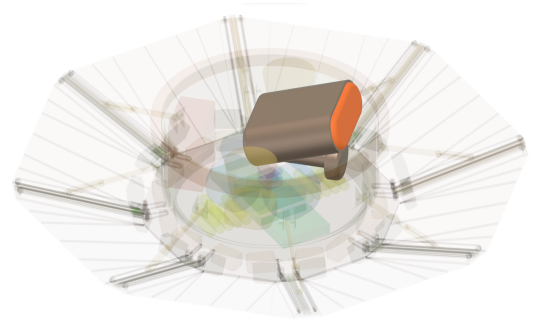


Figure 2: Parachute deployment concept

The deployment system has been placed in this way to satisfy the condition to deploy in any orientation. The demonstrator is expected to behave in one of three ways; it may be stable upright, stable inverted, or completely unstable (tumbling). Figure 3 illustrates how the system will cope with a variety of orientations. The dashed red line represents a danger region, comprised of the wake of the demonstrator and the area directly underneath the demonstrator. If the parachute gets caught in the wake it may not deploy properly, and if it deploys directly into the oncoming airflow under the demonstrator it risks making contact with the demonstrator resulting in tangling and tearing of lines and parachute fabric.

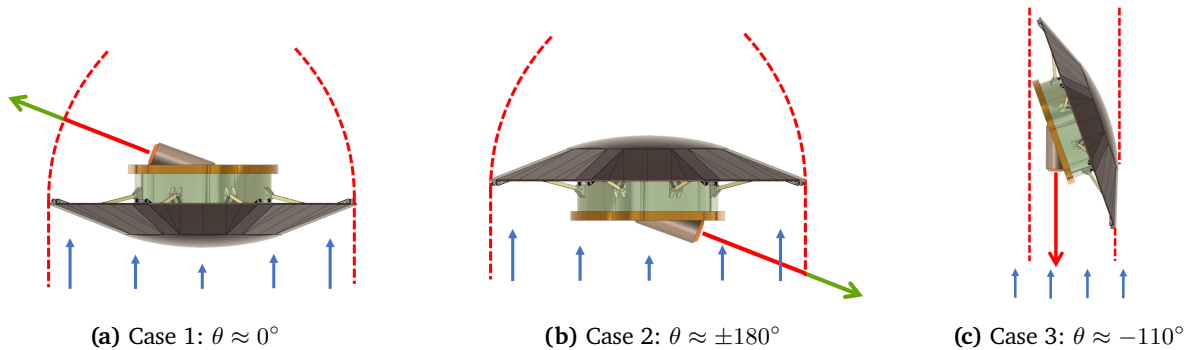


Figure 3: Deployment orientation cases

In Case 1, when the parachute is ejected, it will clear the wake and be exposed to free-stream flow relatively quickly. This will allow the parachute to be blown upwards by the free-stream and begin its inflation vertically above the demonstrator. In Case 2, provided the mortar gives sufficient velocity to the parachute, it will clear the danger region underneath the demonstrator and similarly be introduced

to the free-stream, to inflate vertically upwards. However, in this case the parachute line will necessarily make contact with the demonstrator, namely at the edge of the payload section and along the outer rim of the panels. This condition provides the basis for a significant part of the dynamic modelling analysis, discussed later in Section 4.3. As long as the payload structure and panel rims are designed to have smooth edges which will not cut the parachute line, this should not pose a significant risk to the success of deploying the parachute.

Case 3 is an unlikely edge-case, where the pitch angle (θ) at the time of deployment is -110° and the roll angle is such that the parachute is pointing vertically downwards. In this orientation, deploying the parachute may result in a reduced deployment velocity due to the oncoming airflow, and the parachute may not be able to clear the danger region before making recontact with the demonstrator and becoming tangled. However, this is expected to be only a transient orientation, as the geometry of the demonstrator makes stability at this orientation unlikely. Possible mitigation methods for this case should still be considered, for example implementing a condition set on the flight computers to avoid deploying if the roll and pitch angles have placed the demonstrator in this orientation, and command a brief pause until the orientation has changed. Alternatively, the moving masses that have been used for guidance and control [5] can be displaced before parachute deployment to move the CG radially away from the deployment side (upwards and to the left in Figure 3c), further destabilising this orientation.

4 3-DOF Dynamic Model

With the implementation method and parachute properties broadly selected, a dynamic model was constructed in MATLAB's Simulink environment [16] [17]. This model was developed to simulate three degrees of freedom of the demonstrator after parachute deployment, namely its altitude (x), range (y), and pitch angle (θ). The model is planar, and does not account for roll in the longitudinal axis, in order to simplify the analysis within the project time-frame.

4.1 Level A - Basic Model

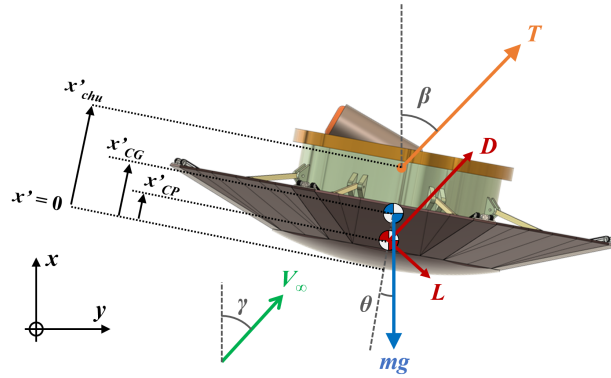


Figure 4: Free-body diagram and axis convention for Level A dynamic model

Figure 4 shows the major forces acting on the demonstrator during its descent under a parachute. Lift (L) and drag (D) act perpendicular and parallel to the free-stream velocity respectively, which is inclined at an angle γ to the Earth-fixed vertical x -axis, and the tension in the parachute line (T) acts at an angle β to the x -axis. It is assumed that β and γ are equal to one another at all times during the model time-frame. This is certainly justifiable in the later stages of descent, where parachute oscillations are expected to be small as described in Table 1, so it will remain approximately aligned with the free-stream. In the early stages of parachute deployment, however, and particularly in the case of inverted deployment, $\beta \neq \gamma$. It is expected that the parachute will not begin opening before line-stretch (commonly referred to as "snatch") takes place, and due to the high descent rate as compared to the ejection speed, the parachute is expected to be blown upwards and away from the demonstrator and

be approximately aligned with γ at the time of line-stretch (the video in Reference [18] of the SpaceX Crew Dragon capsule tumbling before pilot chute deployment illustrates this point well). Therefore in order to keep the assumption of $\beta = \gamma$ valid, the model start time is defined to be an instant after line-stretch, when the parachute begins being pulled out of its bag and inflating. Parachute behaviour after line-stretch is modelled in more detail in the Level B model (Section 4.2); for the Level A model it is assumed to inflate instantaneously.

The aerodynamic forces L , D , and T are defined as follows:

$$L = \frac{1}{2}\rho(V_x^2 + V_y^2)\pi\frac{d_h^2}{4}C_L \quad \text{where } C_L = \begin{cases} \frac{\partial C_L}{\partial \alpha} \cdot \alpha & \text{if } |\alpha| < 20^\circ, \\ \pm \frac{\partial C_L}{\partial \alpha} \cdot 20 & \text{otherwise.} \end{cases} \quad (2)$$

$$D = \frac{1}{2}\rho(V_x^2 + V_y^2)\pi\frac{d_h^2}{4}C_D \quad \text{where } C_D = \frac{1}{2}[(C_{D_0} - C_{D_{90}})\cos(2\alpha) + (C_{D_0} + C_{D_{90}})] \quad (3)$$

$$T = \frac{1}{2}\rho(V_x^2 + V_y^2)\pi\frac{d_{chu}^2}{4}C_{D_{chu}} \quad \text{where } C_{D_{chu}} = 0.55 \quad (\text{see Section 3.1}) \quad (4)$$

The lift coefficient is set to be linear within the range $-20^\circ < \alpha < 20^\circ$ (where $\alpha = \beta - \theta$), and constant outside that range. While this is an over-simplification of the true lift forces acting on the demonstrator, it captures the typical linear lift behaviour near $\alpha = 0^\circ$ and the flat-line region prevents post-stall discontinuities or lift wind-up for large α . Creating a more detailed lift curve with any reasonable level of confidence would require subsonic aerodynamic analysis, which is beyond the scope of this report or the reports of those working on aerodynamic analysis. A value for $\partial C_L / \partial \alpha$ of $1/75 \text{ deg}^{-1}$ was selected.

The drag model enforces two key values, namely that $C_D(0^\circ \pm 180n) = C_{D_0}$ and $C_D(90^\circ \pm 180n) = C_{D_{90}}$, with a smooth cosine interpolation between these points. This captures the typical parabolic behaviour of C_D near $\alpha = 0^\circ$ (as the cosine function approaches a parabola), and enforces a minimum C_D when the demonstrator is on its side. It also implies that the drag coefficient will be similar if the demonstrator is upright or inverted; the upright C_{D_0} at supersonic Mach numbers is estimated to be 1.2-1.4 [19], therefore a slightly lower C_D of 1.1 is used for this model. The inverted C_D is hard to model due to the complex aft-body geometry, however the demonstrator can be idealised as being similar to a flat disk or hollow hemisphere. These have drag coefficients of 1.1 and 1.42 respectively [20], therefore retaining the drag coefficient of 1.1 is a reasonable assumption. The value of $C_{D_{90}}$ was estimated to be 0.4.

It is worth noting with both the lift and drag coefficients that, despite their inaccuracies far from $\alpha = 0^\circ$, the magnitudes of their forces and torques are significantly lower than those produced by the parachute and inertial forces, so the model will not be affected significantly by inaccurate aerodynamic coefficients. At low angles of attack, where the aerodynamic forces are more significant to the response of the demonstrator, accuracy is preserved in the model.

By summing the horizontal and vertical components of forces and equating them to the respective accelerations, and summing moments about the centre of pressure, the equations of motion can therefore be written as follows:

$$\ddot{x} = \frac{1}{m} [(T + D) \cos \beta - L \sin \beta] - g \quad (5)$$

$$\ddot{y} = \frac{1}{m} [(T + D) \sin \beta + L \cos \beta] \quad (6)$$

$$\ddot{\theta} = \frac{1}{I_{z'z'}} [mg \sin \theta (x'_{CG} - x'_{CP}) + T \sin \alpha (x'_{chu} - x'_{CP})] \quad (7)$$

These equations were implemented in a Simulink model. Figure 5 shows the first 20 seconds of the response of the x - and y - velocities (V_x and V_y), and the pitch angle (θ), to an arbitrary set of initial conditions: $V_{\infty 0} = 40$ m/s, $\beta_0 = 10^\circ$, $\theta_0 = 30^\circ$, and $x_0 = 1000$ m.

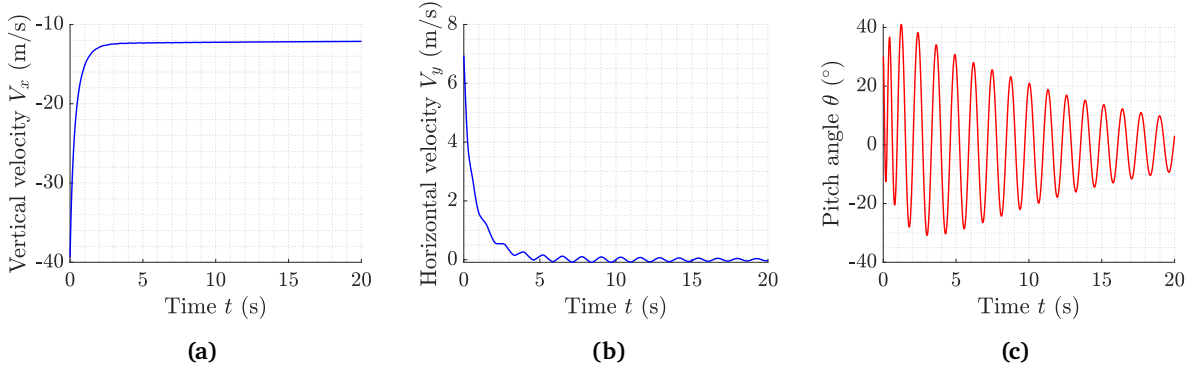


Figure 5: Dynamic response of the demonstrator in Level A model

The response shows a highly oscillatory pitch angle which is lightly damped, and the horizontal velocity tends to zero as the demonstrator slows and approaches a vertical descent. The vertical velocity tends towards -12 m/s which was the design ground impact velocity as specified in Section 3.1. These plots help verify that the model is performing as expected.

4.2 Level B - Parachute Inflation Dynamics

The Level A model assumed instantaneous inflation of the parachute. This gives rise to unrealistically high line tension at $t = 0$, as shown in Figure 6, leading to peak loads $10\times$ greater than the steady load. In order to improve the accuracy of the model, the dynamics of parachute inflation were considered.

A number of analytical and semi-empirical models exist for quantifying the inflation process. Common approaches are based on the conservation of momentum in the parachute radial direction, such as those proposed by Wolf [21] and Macha [22], or mass continuity considering the mass of air within and surrounding the parachute as used by Ludtke [23] and Knacke [10]. The latter of these was briefly explored, and its feasibility to be implemented into a Simulink model assessed. However it was found to require very approximate estimation of key properties, such as the apparent mass m_a of air surrounding the parachute during the inflation time. Even simpler mass-flow approaches, where the parachute is treated as a semi-ellipsoid with constant semi-perimeter and varying volume, were also explored and a Simulink model created. It was deemed to be unsuitable for implementation into the broader 3-DOF dynamic model, however, as the solutions were computationally expensive (complicated by indeterminate properties of ellipsoids) and differed greatly from experimental data. Given the short time-frame of the project, therefore, it was decided to use simple empirical relations proposed by Knacke [10] relevant to ribbon parachutes. He proposed that the opening time t_f (which is the time between initial line-stretch and full inflation) can be approximated via the relation:

$$t_f = \frac{n \cdot d_{chu}}{V^{0.9}} \quad \text{where } n = 8 \text{ for a ribbon parachute, } d_{chu} \text{ is in ft, and } V \text{ is in ft/s} \quad (8)$$

During this time, the area and drag coefficient are evolving together. Experimental testing has shown that they can be treated as a single property $C_D S$ which, for ribbon parachutes, tends to increase linearly for $0 < t < t_f$ (see Figure 5-38 in Reference [10]). This is then easily implemented as a ramp in Simulink, whose slope is S/t_f , applied to the parachute area, and a constant drag coefficient such that $C_D S$ varies linearly until the parachute area is $\frac{\pi d_{chu}^2}{4}$.

The resulting force-time plot for the Level B model is given in Figure 6, and shows a much lower peak tension compared to the instantaneous opening of the Level A model.

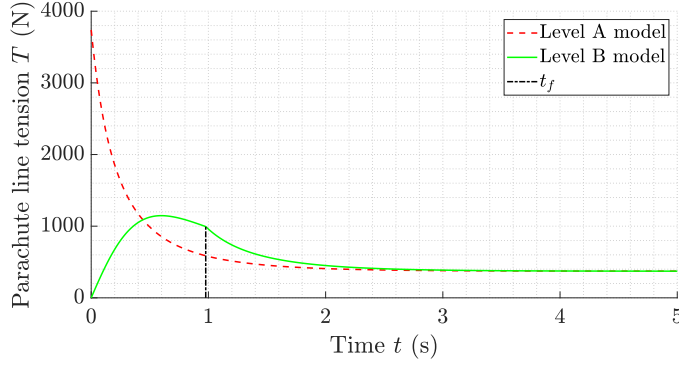


Figure 6: Parachute loads with opening dynamics modelled

4.3 Level C - Line-Body Interactions

In the Level C model, parachute opening dynamics from Level B were retained and a further contribution from the parachute lines was considered.

Since the demonstrator may be tumbling or falling in any orientation before the parachute is deployed, the model should be able to account for the parachute lines making contact with the demonstrator's geometry. There exists an angle of attack α_1 above which the parachute line will first come into contact with the edge of the payload fairing. As α increases above this critical value, the parachute tension vector T acting at the point x'_{chu} will no longer be inclined at an angle β to the vertical; instead it will remain at a constant angle α_1 from the body-fixed x' -axis, and its angle to the Earth-fixed x -axis will be $\theta + \alpha_1$. The edge of the payload fairing is modelled as a smooth surface, therefore the rest of the parachute line does not exert any additional tensile force on the demonstrator. Instead, there will be a smaller force T_1 which bisects the angle created by the parachute line, acting on the edge of the payload fairing. These forces are illustrated in Figure 7a, and a new angle ζ is defined as the angle between T_1 and the body-fixed x' -axis.

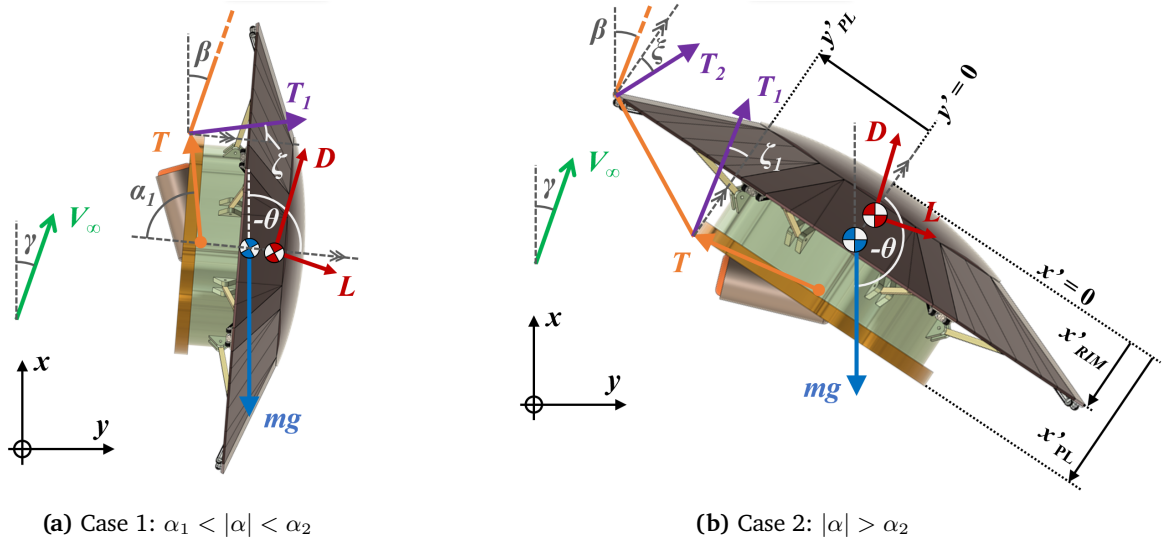


Figure 7: Free-body diagrams and axis convention for Level B dynamic model

The introduction of an extra force T_1 and the new specification of the direction that T points in gives rise to new equations of motion, valid for $\alpha_1 < |\alpha| < \alpha_2$ (note the $\pm/\mp$ ambiguity arises from the sign of α ; in all cases the upper sign should be used for positive α and the lower sign for negative α):

$$\ddot{x} = \frac{1}{m} [D \cos \beta - L \sin \beta + T \cos (\theta \pm \alpha_1) - T_1 \cos (\theta \mp \zeta)] - g \quad (9)$$

$$\ddot{y} = \frac{1}{m} [D \sin \beta + L \cos \beta + T \sin (\theta \pm \alpha_1) + T_1 \sin (\theta \mp \zeta)] \quad (10)$$

$$\ddot{\theta} = \frac{1}{I_{z'z'}} \left[mg \sin \theta (x'_{CG} - x'_{CP}) \pm T \sin \alpha_1 (x'_{chu} - x'_{CP}) \pm T_1 [\cos \zeta (y'_{PL}) + \sin \zeta (x'_{PL} - x'_{CP})] \right] \quad (11)$$

As α increases further, there will be a second critical angle of attack α_2 where there is an additional point of contact between the parachute lines and the rim of the panels. In a similar way, this gives rise to a new force T_2 acting on the demonstrator at the rim, bisecting the parachute line angle, at an angle ξ to the x' -axis. Furthermore, ζ will become fixed at ζ_1 for any further increase in α . These forces are illustrated in Figure 7b and the relevant equations of motion for $|\alpha| > \alpha_2$ are as follows:

$$\ddot{x} = \frac{1}{m} [D \cos \beta - L \sin \beta + T \cos (\theta \pm \alpha_1) - T_1 \cos (\theta \mp \zeta_1) + T_2 \cos (\theta \pm \xi)] - g \quad (12)$$

$$\ddot{y} = \frac{1}{m} [D \sin \beta + L \cos \beta + T \sin (\theta \pm \alpha_1) + T_1 \sin (\theta \mp \zeta_1) + T_2 \sin (\theta \pm \xi)] \quad (13)$$

$$\ddot{\theta} = \frac{1}{I_{z'z'}} \left[mg \sin \theta (x'_{CG} - x'_{CP}) \pm T \sin \alpha_1 (x'_{chu} - x'_{CP}) \pm T_1 [\cos \zeta_1 (y'_{PL}) + \sin \zeta_1 (x'_{PL} - x'_{CP})] \right. \\ \left. \pm T_2 [\cos \xi (\frac{d_h}{2}) - \sin \xi (x'_{RIM} - x'_{CP})] \right] \quad (14)$$

For a more detailed derivation of the relevant angles and magnitudes of T_1 and T_2 , see Appendix C.

With the equations of motion now specified for all pitch angles, the model was run and analysed at the extremes of deployment conditions. Figure 8 shows the first 20 s of the dynamic response of the demonstrator to a set of initial conditions which ensures line contact ($V_{\infty 0} = 40$ m/s, $\beta_0 = 10^\circ$, $\theta_0 = 160^\circ$ (near-inverted), and $x_0 = 1000$ m). The response shape is generally similar to that of the Level A model (Figure 5), but shows a greater initial level of damping to the high pitch angles during body-line contact, and a gentler initial V_x reduction due to the Level B parachute opening dynamics.

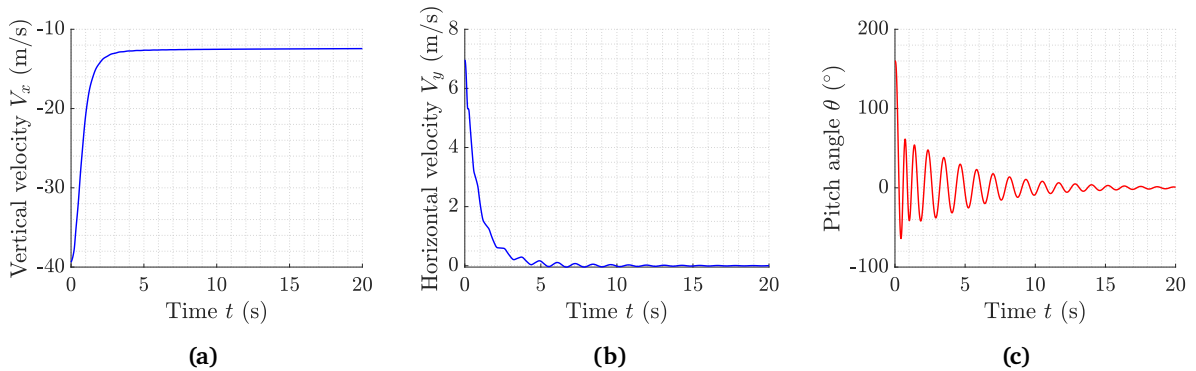


Figure 8: Dynamic response of the demonstrator in Level C model

Figure 9 shows a comparison of the total moment produced by the parachute line in the Level B (force T only) and C (forces T , T_1 , and T_2) models, both run with the same initial conditions as above, in order to further illustrate the difference in the loads and response with the new extended equations of motion.

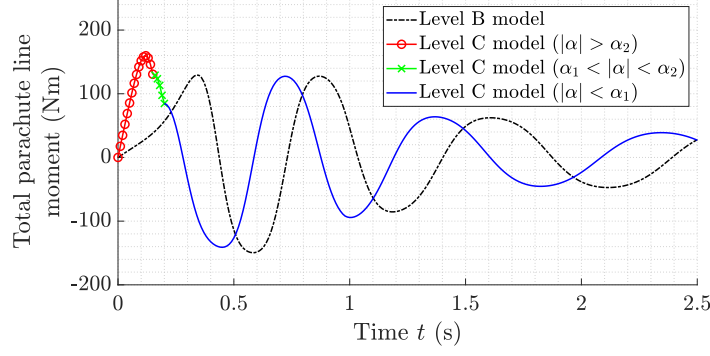


Figure 9: Comparison of moments imparted by parachute line forces in Level B and C models

Parts of the graph where the model is subject to different equations of motion have been coloured differently for clarity. The plot illustrates that the inclusion of extra forces due to the parachute lines interacting with the demonstrator increase the total moment during the first few ms of parachute deployment, as compared to the level B model. It also indicates that the lines only interact with the demonstrator during the initial reorientation i.e. the angular momentum of the demonstrator is too small for it to recontact the lines after it passes through its first half-oscillation. This matches physical intuition, that the initial reorientation event is the most intense and subsequent oscillations are smaller.

5 Results and Load Envelopes

With the model producing sensible results for a range of initial conditions, it is possible to extract key data on the loads that the demonstrator experiences during parachute deployment. The longitudinal and lateral inertial forces (in body-fixed x' - and y' - coordinates) can be determined as follows:

$$a_{x'} = \frac{1}{g} [\ddot{x} \cos \theta + \ddot{y} \sin \theta + g] \quad (15)$$

$$a_{y'} = \frac{1}{g} [-\ddot{x} \sin \theta + \ddot{y} \cos \theta] \quad (16)$$

The parachute line tension and angular inertial forces can be directly extracted from the Simulink model. Together, a complete account of the loads experienced by the demonstrator can be plotted, as shown in Figure 10 with the same initial conditions as in Figures 8 and 9.

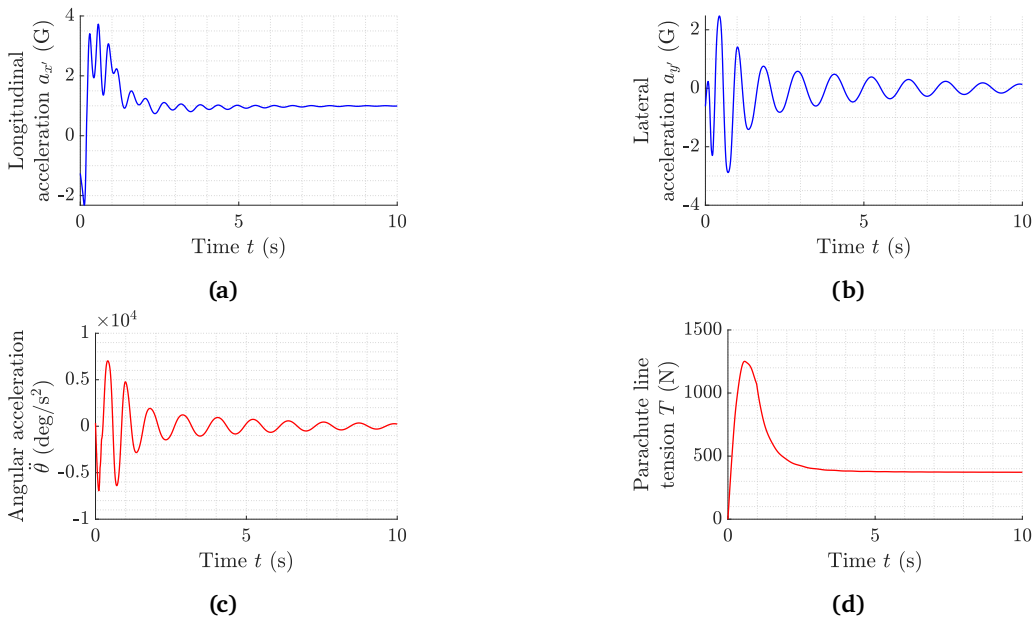


Figure 10: Load output from Level C dynamic model

From these data, the maxima of loading can be inferred for any given set of initial conditions. Therefore, by sweeping the initial conditions across a range of values, a load envelope can be formed to assess all possible deployment cases. It is assumed that in the terminal portion of descent, the flight-path angle γ will be near-vertical [4], therefore β_0 has been kept at zero. To reduce the number of initial condition sweeps required, it is also assumed that the demonstrator will be at terminal velocity at all times during this final phase of descent. For simplicity, the higher value of $C_{D_h} = 1.1$ has been assumed, as the demonstrator is unlikely to be falling on its side where $C_{D_{90}} = 0.4$ for any significant period of time. The terminal velocity can therefore be set as the initial velocity condition for the model, based on a given initial altitude, via the following relation:

$$V_{\infty 0} = \sqrt{\frac{8m_h g}{\rho(x)\pi d_h^2 C_{D_h}}} \quad \text{where } \rho(x) \text{ is found using MATLAB's } \text{atmosisa}() \text{ function} \quad (17)$$

The model was then swept over a range of initial values for altitude x_0 and pitch angle θ_0 , between 100 & 2000 m, and 0° & 180° respectively. Negative pitch angles have not been considered as the 3-DOF model assumes a symmetric demonstrator (and the dynamics of mortar deployment, which are highly dependent on the sign of θ_0 , have not been modelled). An upper limit of 2000 m has been selected as this corresponds to nearly 3 minutes of descent under a parachute; a higher deployment altitude would cause unnecessary drift downwind from the desired landing zone. Additionally, a lower deployment altitude allows for greater time spent in phase C of the mission, where key data about the dynamic behaviour at low speeds without a parachute can be recorded. Figure 11 shows the maximum absolute values of each of the four loading metrics across these initial conditions.

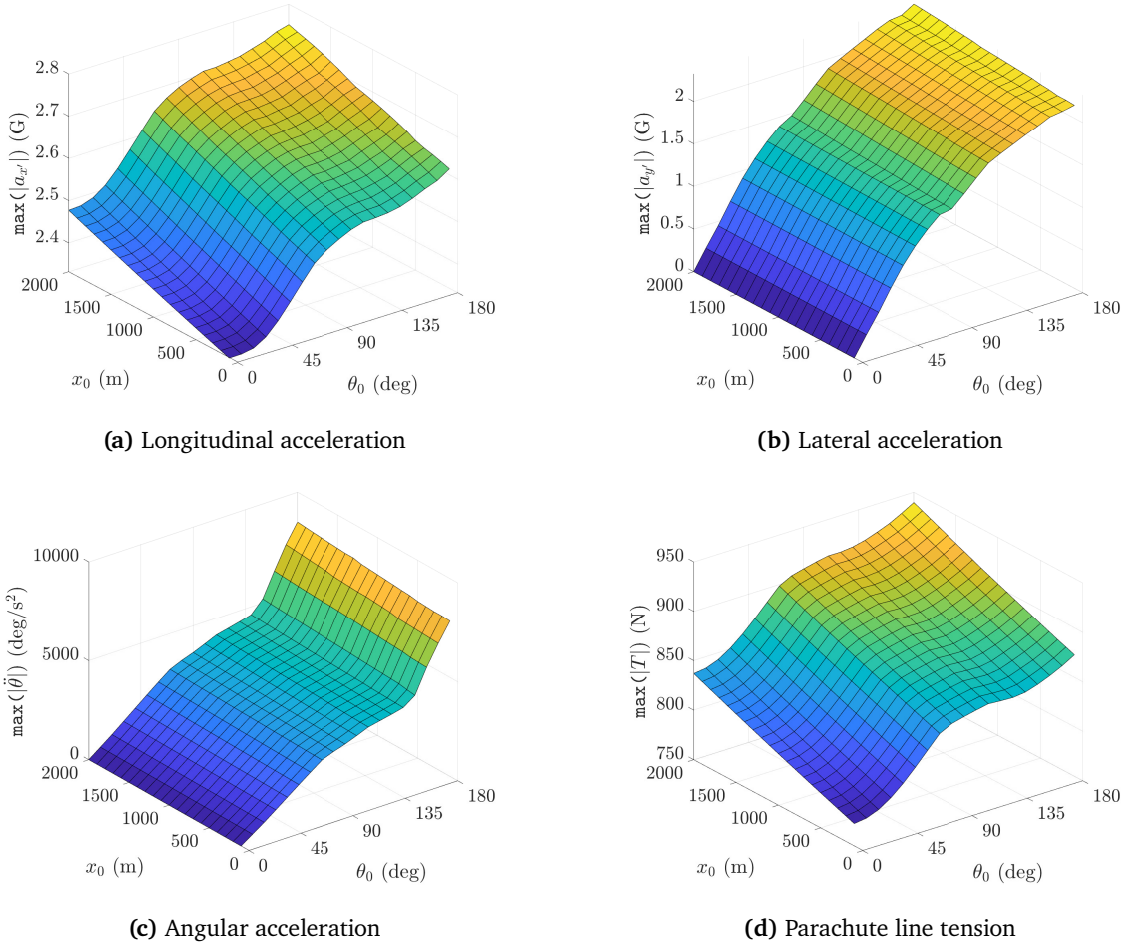


Figure 11: Loading maxima across initial condition array of $100 \leq x_0 \leq 2000$ and $0^\circ \leq \theta_0 \leq 180^\circ$

These results can give insight into which loads are constraining, and their sensitivity on the input conditions. For example, longitudinal and lateral loads are small compared to the expected quasi-static loading due to vibrations during launch [24] and the high deceleration during reentry [4], therefore these are not particularly constraining loads. The parachute line tension is a concentrated point force acting at the connection between the parachute lines and the payload structure, therefore the connecting hardware needs to be designed to meet the peak loading requirements so this will be a constraining force. The maximum angular acceleration $\ddot{\theta}$ is a key indicator of how sudden the reorientation manoeuvre is, and is overwhelmingly dependent on θ_0 rather than altitude as illustrated in Figure 11c. While few of the studies from other subgroups have considered angular loads, a simple calculation shows that an angular acceleration of 8400 deg/s^2 (as experienced when $\theta_0 = 0^\circ$ at 2000 m) would cause a 7.7 G equivalent linear acceleration load at the edge of the demonstrator panels, which is of similar magnitude to launch and reentry loads. A +7.7 G load at one edge of the demonstrator coupled with a -7.7 G load at the other may produce torques which are too high for the structure to withstand. Until further structural analysis is done in future design iterations, a limit on angular acceleration cannot be justifiably set, however this model would prove to be useful in creating safe deployment envelopes based on the maximum allowable loads.

For example, if an arbitrary maximum tip acceleration of $\pm 3.5 \text{ G}$ is enforced (equivalent to 4100 deg/s^2), with a maximum parachute line tension of 900 N, a deployment envelope can be constructed as shown in Figure 12. An additional constraint could be set that the parachutes should be opened above 250 m altitude, in order to ensure at least 20 seconds nominally of descent under a parachute (otherwise short delays in parachute inflation below this altitude would significantly increase the risk of a full-speed ground impact).

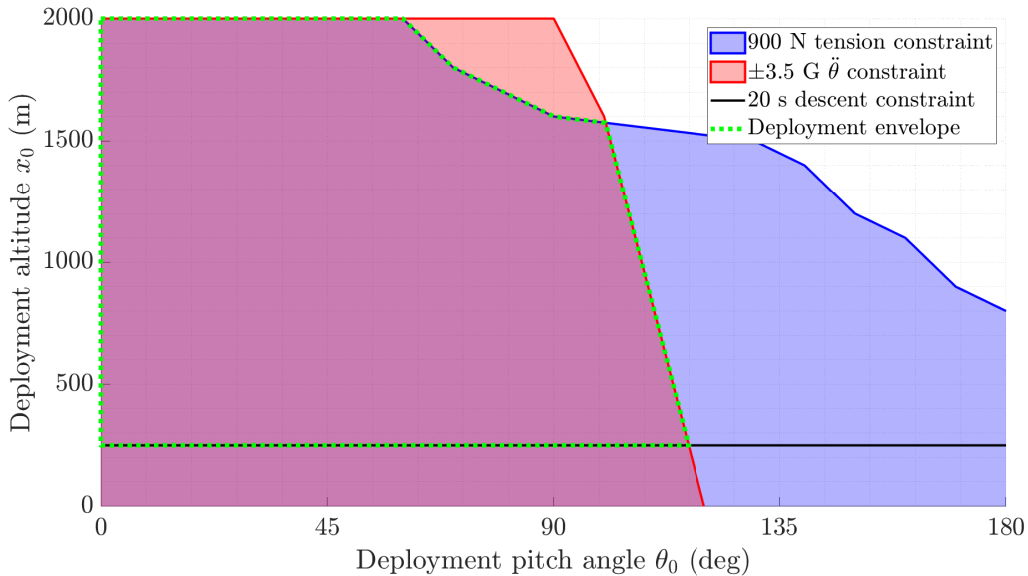


Figure 12: Example of altitude & pitch angle deployment envelope

This is an unconventional loading envelope, as they are typically plotted with axes of altitude and speed/Mach number. However it captures the fact that the pitch angle at deployment has a significant impact on the resultant loads experienced by the demonstrator. Interpreting the envelope in Figure 12 would imply that the demonstrator should never be deployed if it has a pitch angle more than 120° for this given set of structural constraints. This would imply that the recovery system would not be robust against deploying in any orientation, which was one of the main aims for this system. Therefore the tip acceleration constraint could be relaxed (e.g. by strengthening the structure where necessary), or the parachute line mounting point x'_{chu} lowered to reduce moments on the demonstrator and resulting angular accelerations.

6 Discussion and Improvements

The 3-DOF model described in Section 4 has been shown to produce a dynamic response that satisfactorily matches the expected behaviour, and useful loading information can be extracted to construct preliminary deployment envelopes. The focus should be on the development of the model as a generalised tool for analysing the terminal portion of the Space HAVEN mission; many properties and constraints are still unknown at this level of design maturity, but the model will be a useful tool in moving forwards with the design to ensure the recovery system performs within structural limitations and is well-integrated into the mission as a whole.

There are, however, many areas in which the model can be improved in order to better capture the response of the demonstrator during recovery. One complex but necessary improvement would be extending the model to 6 degrees of freedom, primarily to capture yaw and roll responses. The mortar deployment direction varies significantly with yaw and roll, therefore these states would need to be considered in order to more accurately quantify the demonstrator's motion and obtain more detailed loading data.

Even remaining within a 3-DOF space, the model could be further improved by the addition of mortar dynamic modelling. The period before line-stretch (where this model's time starts at $t = 0$) would still see forces exerted on the demonstrator in a physical world, and particularly in the case of a tumbling demonstrator, the pitch angle may change between mortar ignition and line-stretch. The assumption of $\beta = \gamma$ would therefore need to be discarded, and the parachute inside its bag prior to line-stretch could be modelled as a rigid body attached to the demonstrator by the parachute lines and subject to aerodynamic forces in the free-stream. Snatch loads at line-stretch can then be inferred.

Features which are already implemented in the model but which would benefit from a more rigorous analysis include the parachute opening dynamics, and all subsonic aerodynamic force coefficients. These have been estimated for the purpose of this report, but should ideally be obtained from methods such as panel code, vortex lattice, or even CFD simulations, and ultimately verified with wind tunnel testing. Dynamic force coefficients should also be considered, such as pitch rate damping coefficients C_{M_q} . These have been explored for HATHOR at supersonic speeds [25], but a subsonic analysis would aid the accuracy of post-parachute inflation oscillation damping as well as pre-deployment dynamics. It may be able to provide insight as to whether the demonstrator is expected to remain upright or inverted at subsonic Mach numbers, which may allow a more refined parachute deployment system design to be optimised for deployment at a specific orientation. This would then facilitate a more detailed design of the parachute and mortar deployment mechanism, in order to prepare the demonstrator for its test-flight in the Space HAVEN mission.

7 Conclusions

This report detailed the high level design and implementation of a parachute recovery system for a foldable aeroshell demonstrator. The proposed deployment system is expected to be able to deploy a parachute at almost any orientation without a risk of line entanglement or recontact with the demonstrator. The model developed captures a number of key properties of the dynamics of parachute deployment, including simple parachute inflation dynamics and line-body interactions at high pitch angles. Loading information can be extracted from the dynamic response, and the model is capable of providing the necessary data for construction of an altitude-pitch angle parachute deployment envelope, subject to structural constraints. The model provides a good starting point for the analysis of what is a very chaotic and violent portion of descent, and ways to improve the accuracy and validity have been discussed including extension to 6 degrees of freedom. Additional work carried out during the project has been presented in the appendices, including mass and power budgeting and CAD work to aid with estimating inertial properties and validating panel geometries.

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Appendix

A Subsystem Breakdown, Mass and Power Budgets

A list of subsystems and their expected masses is given in Table 2. In most cases, the masses have been estimated by finding products online that satisfy the requirements for this mission, and specifics on these systems and their properties can be found in the relevant reports of the FAD06 sub-team [2] [5] [6]. In cases where this information is less easily accessible, best estimates have been made based on comparison to other payloads, small satellites, and deployable reentry vehicles. All masses are collated here for ease of reference.

Table 2: Mass and power breakdown of subsystems and components

Subsystem/Component	Quantity	Unit mass estimate (kg)	Line total mass (kg)	Unit power estimate (W)	On-time (s)	Line total energy (Wh)
Deployment actuators	8	0.3	2.4	3	15	0.1
Parachute	1	1.2	1.2	-	-	-
Parachute mortar and ignition	1	1	1	10	1	0.003
Moving masses	8	1.5	12	-	-	-
Mass movement hardware	8	0.2	1.6	-	-	-
Mass movement actuators	16	0.2	3.2	4	50	0.9
Reaction wheels	4	0.8	3.2	12	360	4.8
Clamp-band hardware [26]	1	0.8	0.8	-	-	-
Combined IMU/GPS unit	3	0.02	0.06	0.4	3600	1.2
Star sensor	1	0.35	0.35	0.5	360	0.05
Sun sensor	1	0.1	0.1	0.6	360	0.06
Contact heat sensor	16	0.01	0.16	-	-	-
On-board computer	2	0.25	0.5	7	3600	14
Data handling and storage unit	2	0.3	0.6	1.5	3600	3
Telemetry system	1	1.5	1.5	20	3600	20
Battery pack*	1	0.9	0.9			
Total subsystem mass (kg):			29.6	Total energy (Wh):		44
Accounted structural mass (kg):			5 to 10			

The upper bound for the sum of the subsystem mass and the accounted structural mass is approximately 39.6 kg. There is a significant portion of unaccounted mass, such as that of general fastening hardware, hinges, wiring, and strengthening structure for the parachute mount point. This is estimated to be on the order of 15% of the accounted mass, bringing the total mass of the demonstrator to ≈ 45 kg. Additionally, it is expected that the mass will increase as future iterations of the design are completed. Assuming the project is nearing completion of a preliminary design, based on historic trends the mass would on average increase by a further 15% before project completion [27].

*The batteries have been sized based on the estimated energy and power requirements. The standard 18650 lithium-ion cell is used as a baseline for building up a battery pack [28]; with a nominal voltage of 3.6 V, nine of these would be required in series to produce a desired output voltage of 30 V (as required by the actuators). At 2.85 Ah per cell, this would give a total energy capacity of 86 Wh, which satisfies the requirement for 44 Wh as per Table 2. Since telemetry and data storage are such a key part of recording this mission and reconstructing the trajectory after landing, however, the total battery capacity should be increased for extra safety margin. This will mean a greater allowable pre-launch on-time, and a longer recovery search time before battery depletion should the demonstrator land far from the expected launch site. Therefore 18 cells will be used in a 9S2P configuration, at 48 g per cell, for a total battery mass of 0.864 kg (rounded to 0.9 kg in Table 2), and an energy capacity of 172 Wh. Furthermore, the maximum discharge rate per cell is 1 C, so a 9S2P configuration will be able to discharge at a maximum current of $(2.85 \text{ Ah} \times 1 \text{ C} \times 2 = 5.7 \text{ A})$. At a 30 V output voltage, this is a 171 W maximum continuous power output. This is sufficient to power all systems on the demonstrator, as the peak power requirement is expected to be < 100 W, occurring during Phase A of the flight (see Figure

1) where the reaction wheels and deployment actuators will both be operating alongside all computer and telemetry systems.

B CAD Work - Inertial Tensor Estimation and Folding Patterns

Throughout the project, it was necessary to have an up-to-date CAD model to better visualise the demonstrator and verify design decisions. Alongside parachute modelling, the CAD was produced and updated at the request of other members in order to benefit the whole team. It enabled placement of components and subsystems within the payload bay, and estimation of the inertia tensor without requiring tedious hand calculations. As with mass and power estimates, volumes and dimensions of components were taken from datasheets of appropriate products online. Figure 13 shows how the subsystems and components have been nested in the payload bay in order to maximise volume efficiency and maintain a low CG for stability during the guided portion of flight [5]. Bright colours have been used for ease of distinguishing components, and do not reflect the actual materials or appearances.

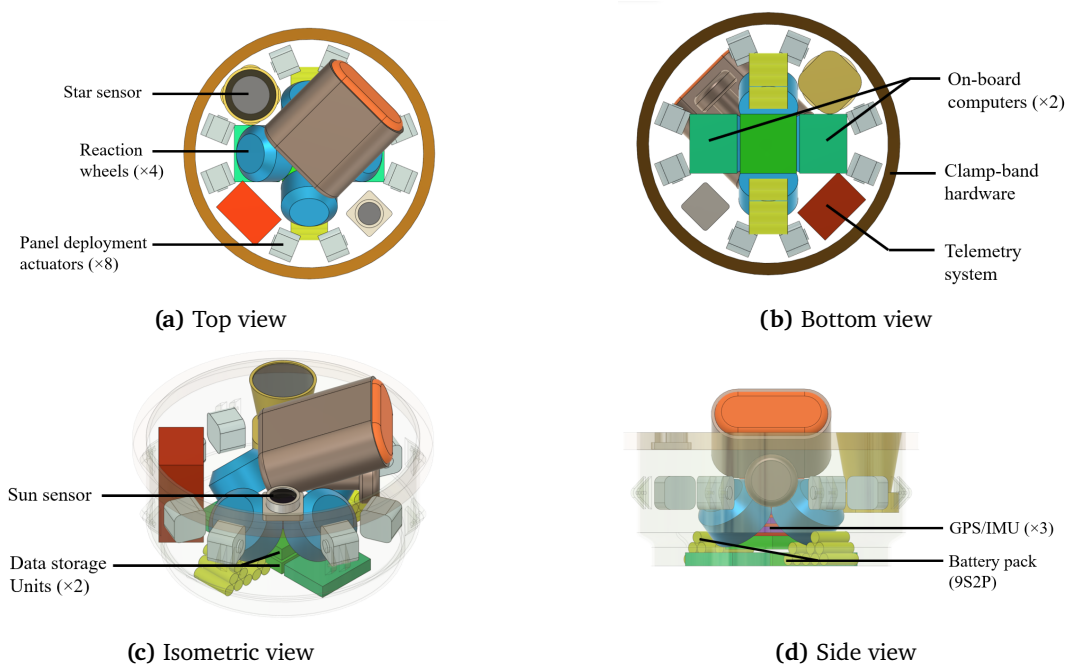


Figure 13: Multi-view of payload bay to illustrate subsystem placement

This payload is shown in context to the rest of the demonstrator in Figure 14. The large actuators required for the moving mass control system have been placed outside the payload bay due to tight volume constraints. It should be noted that the payload internal structure has not been designed; it simply represents a bounding cylinder within which to place all subsystems.

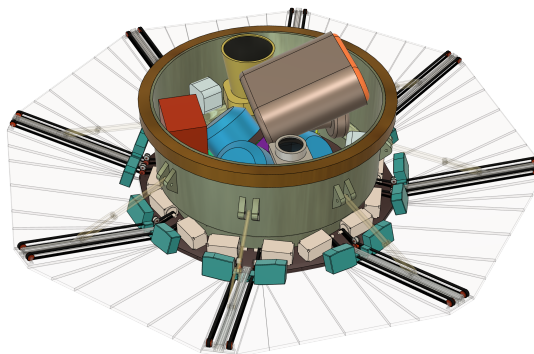


Figure 14: Payload bay in context with demonstrator structure

By correctly inputting the mass of each component, an estimation of the moments of inertia and CG location could be extracted from the CAD model. Components were initially placed in a configuration that aimed to minimise the lateral distance of the CG from the vertical axis. After placing components, the CG was found to be < 1 mm laterally from the vertical axis of the demonstrator, which is small enough to neglect asymmetry at this design stage. The inertial properties are summarised in Table 3.

Table 3: Summary of inertial properties of demonstrator in stowed and deployed configurations

	stowed	deployed
x'_{CG} (mm)	117	103
$I_{x'x'}$ (kgm ²)	1.38	1.80
$I_{y'y'}$ (kgm ²)	0.83	1.02
$I_{z'z'}$ (kgm ²)	0.84	1.02

The cross-products of inertia, such as $I_{x'z'}$ and $I_{x'y'}$, are all at least two orders of magnitude lower than those in Table 3, therefore they can be reasonably neglected for the sake of simplifying control wheel analysis [2], moving mass control analysis [5], and parachute dynamic analysis.

The panel, rib, and arm geometry were included in this CAD model to aid the demonstrator structural design sub-team (FAD05) in verifying that the arm geometry was able to actuate the ribs, and that the panel origami pattern was valid. Figure 15 shows stills from a video animation created for this purpose.

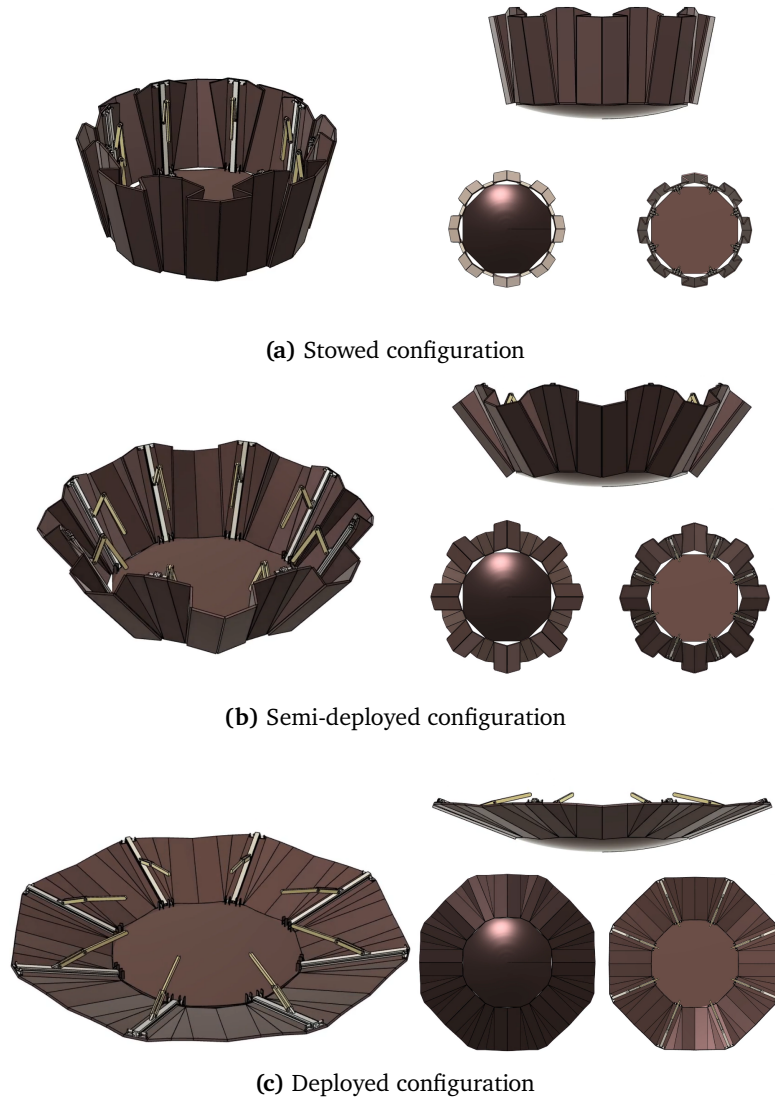


Figure 15: Multi-view illustration of panel folding patterns

C Parachute Line Geometry Derivations

A detailed derivation of the angles and forces necessary to fully define the equations of motion presented in Section 4.3 (Equations 9 to 14) is presented below.

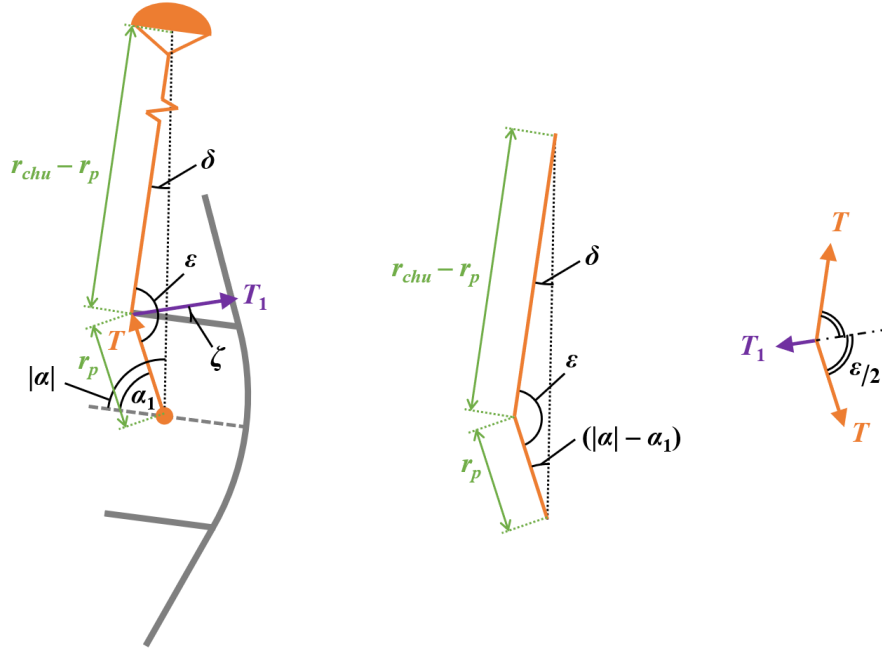


Figure 16: Diagrams for derivation of Level C Case 1 equations of motion

Using the sine rule on the extracted triangle in Figure 16:

$$\frac{\sin(\delta(\alpha))}{r_p} = \frac{\sin(|\alpha| - \alpha_1)}{r_{chu} - r_p} \Rightarrow \delta(\alpha) = \arcsin \left[\frac{r_p}{r_{chu} - r_p} \sin(|\alpha| - \alpha_1) \right] \quad (18)$$

where α_1 and r_p can be determined from geometric properties using the CAD model as a reference, and r_{chu} is the parachute length, defined to be 8 m in Section 3.1. By summing angles inside the triangle:

$$\varepsilon(\alpha) = 180^\circ - \delta(\alpha) - (|\alpha| - \alpha_1) \quad (19)$$

The angle ζ that the extra parachute force T_1 acts at, relative to the local x' -axis, can be found by the following equation, which assumes T_1 perfectly bisects ε (a consequence of the smooth body assumption):

$$\zeta(\alpha) = \frac{\varepsilon(\alpha)}{2} - \alpha_1 \quad (20)$$

The magnitude of T_1 can therefore be found by balancing the forces shown in the right-hand diagram, giving:

$$T_1(\alpha) = 2T \cos \frac{\varepsilon(\alpha)}{2} \quad (21)$$

T_1 , α_1 , and ζ can thus be input into the equations of motion for Case 1 of the Level C model.

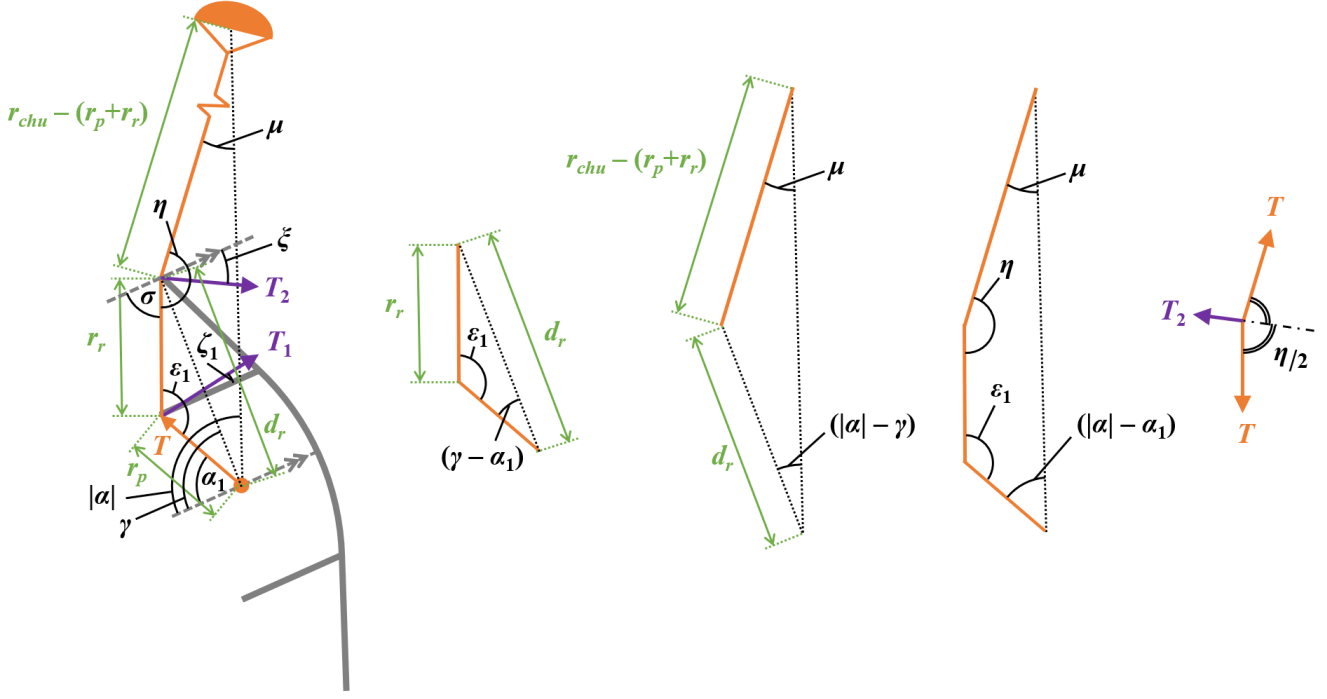


Figure 17: Diagrams for derivation of Level C Case 2 equations of motion

Starting again with the sine rule, applied on the smaller extracted triangle shown in Figure 17:

$$\frac{\sin(\epsilon_1)}{d_r} = \frac{\sin(\gamma - \alpha_1)}{r_r} \Rightarrow \epsilon_1 = \arcsin \left[\frac{d_r}{r_r} \sin(\gamma - \alpha_1) \right] \quad (22)$$

It is worth noting that this constant angle ϵ_1 can be used as the indicator as to whether the line has made contact with the rim of the demonstrator; rather than checking for $|\alpha| > \alpha_2$, where α_2 is difficult to find, it is possible to check for $\epsilon(\alpha) < \epsilon_1$. If this condition is satisfied, then the demonstrator is in Case 2 where both T_1 and T_2 are present. This is implemented in the Simulink model.

The angle ζ_1 that T_1 acts at becomes constant for any angles of attack where the line makes contact with the rim, and is defined as:

$$\zeta_1 = \frac{\epsilon_1}{2} - \alpha_1 \quad (23)$$

Similarly, the magnitude of T_1 does not change for any higher angles of attack, and is fixed at:

$$T_1 = 2T \cos \frac{\epsilon_1}{2} \quad (24)$$

Applying the sine rule on the larger triangle in Figure 17:

$$\frac{\sin(\mu(\alpha))}{d_r} = \frac{\sin(|\alpha| - \gamma)}{r_{chu} - (r_p + r_r)} \Rightarrow \mu(\alpha) = \arcsin \left[\frac{d_r}{r_{chu} - (r_p + r_r)} \sin(|\alpha| - \gamma) \right] \quad (25)$$

Summing the angles in the extracted quadrilateral:

$$\eta(\alpha) = 360^\circ - (\mu(\alpha) + (|\alpha| - \alpha_1) + \epsilon_1) \quad (26)$$

The angle ξ that T_2 acts at relative to the x' -axis can be found by summing the angles along the straight line at the rim (shown on the left-most diagram of Figure 17):

$$\xi(\alpha) = 180^\circ - \sigma - \frac{\eta(\alpha)}{2} \quad (27)$$

and the magnitude of T_2 can be determined with a force balance as shown in the right-hand diagram of Figure 17:

$$T_2(\alpha) = 2T \cos \frac{\eta(\alpha)}{2} \quad (28)$$

T_2 , ξ , ζ_1 and the updated T_1 can thus be input into the equations of motion for Case 2 of the Level C model.